TOPS Hackathon — Problem Statement

Section 1: Problem Statements

1. Faculty Workload & Academic Automation Platform

Problem Statement

Faculty members spend a significant portion of their time on repetitive administrative tasks — attendance management, assignment evaluation, report generation, timetable preparation, student mentoring, accreditation documentation (NAAC/NBA), internal assessments, and parent communication. This leaves little time for quality teaching, research, curriculum development, and personalized student support, and existing academic management systems mostly just record data rather than actively automating it. Build a platform that automates these repetitive academic and administrative tasks so faculty can focus on teaching and mentoring.

Target Users

Schools, colleges, universities, technical institutes, faculty members, academic coordinators, department heads, and school/college management.

Expected Features / Deliverables

- AI Attendance Analytics
- Automated Assignment Evaluation
- AI Question Paper Generator
- Student Performance Dashboard
- Timetable Optimization
- Academic Report Generator
- Parent Communication Module
- Accreditation Documentation Assistant
- Faculty Productivity Dashboard
- Smart Notification & Reminder System

Suggested Tech Stack

Python, Django REST Framework / FastAPI, React.js, PostgreSQL, OpenAI API, Scikit-learn, Docker, Firebase Cloud Messaging.

2. AI-Powered Academic Integrity & Assignment Evaluation Platform

Problem Statement

The widespread use of AI tools and online resources has made it increasingly difficult for institutions to ensure academic integrity. Faculty spend considerable time verifying originality, evaluating coding quality, detecting plagiarism, identifying AI-generated content, and giving personalized feedback — and existing plagiarism checkers only catch copied text, not conceptual understanding or reasoning. Build a platform that evaluates assignments, code, reports, and presentations for originality and quality, and generates useful feedback automatically.

Target Users

Schools, colleges, universities, technical institutes, faculty members, students, examination departments, and accreditation bodies.

Expected Features / Deliverables

- AI Content Detection
- Advanced Plagiarism Detection

- Source Code Quality Analysis
- GitHub Repository Evaluation
- Rubric-based Automatic Assessment
- Personalized Feedback Generator
- AI Viva Question Generator
- Student Improvement Tracker
- Faculty Analytics Dashboard
- Batch Performance Reports

Suggested Tech Stack

Python, FastAPI, React.js, PostgreSQL, OpenAI API, Hugging Face Transformers, CodeBERT, GitHub API, Docker.

Section 2: Requirements & Rules

Team & Eligibility

- Team size: 2–4 members. Cross-batch / cross-track teams are allowed.
- Each team selects ONE problem statement from Section 1 once it is revealed.
- A team may not switch problem statements after the reveal window closes.

Rules

- Problem statements remain sealed and are revealed only at the official start time.
- No pre-built projects or previously written application code. Standard scaffolding/boilerplate (create-react-app, django-admin startproject, etc.) is allowed.
- Teams may use the suggested tech stack for their chosen problem, or propose an equivalent alternative — subject to mentor approval before the midpoint check-in.
- Use of AI coding assistants is permitted for productivity, but the team must be able to explain and defend every part of their submission during judging.
- Plagiarism, reused prior projects, or misrepresenting others' work as original leads to disqualification.

Deliverables

- A GitHub repository (private, with the TOPS evaluator account added as a collaborator).
- A README covering: tech stack used, local setup instructions, and which features were completed.
- A live demo of the working application (screen share) — no slide decks.

Judging Criteria

Criteria	Weight
Functionality (does the core flow work end-to-end?)	30%
Code quality & structure	20%
UI/UX	20%
Problem understanding & relevance of solution	20%
Presentation & demo clarity	10%

Code of Conduct

- Respectful collaboration within and across teams at all times.
- Mentors are available for technical unblocking, not for writing your solution.
- Any dispute on scoring should be raised with the lead evaluator immediately after results, not during judging.